ITECH3102: Assessment Task 4: Capstone Project (50%)

Semester 2, 2026

Due date: Sunday, Oct 18, 2026 - 23:59 (Week 12)

Summary

Task Type: Capstone Project (50%)
Due Date: Sunday, Oct 18, 23:59 (Week 12),
Weighting: 50%

1. Design plan (15%, word count: 800)
2. Implementation of design plan with analysis report (35%, word count: 1200)

Learning Outcomes Assessed: K3, S1, S2, S3, A1, A2, A3

Purpose: Demonstrate students' understanding data visualization through designing and implementing an interactive data visualization, and skills in creating data visualizations for different contexts using a range of techniques and tools.

Requirements: Design and implement an interactive dashboard based on a provided specification. This assessment task involves (1) developing a design plan (proposal) for proposed interactive dashboard, (2) implementing the dashboard by visualization based on the design plan by using software with analysis report.

Assessed by: Lecturer/Tutor

Submission:

- Upload (1) your design plan and (2) implementation visualization with analysis report to Moodle.
- The name of your documents should be your name plus your ID.
- Late penalty: A 5% penalty will be applied to the total mark for each late after the due date (including weekends and public holidays).

Feedback: Marks in fdlGrades and Moodle, individual feedback provided via Moodle

Details of assessment task 4

This is an individual task. The objective of the assessment task 4 is to apply the knowledge, theory, and skills that you have learned this semester about Data Visualization and put it to application of designing an interactive dashboard in a creative effort. Keys to the assessment task 4 are to design a dashboard and implement it, which general users will find your dashboard interesting and intriguing. In addition, you may design scenarios where you clearly identify a set of users or analysts who care deeply about that data you select.

1. Create a plan of your visualization (15%)

The first part of the assessment task 4 is to create a design plan for your interactive dashboard. You first should select one of two datasets provided below and then create a proposal in which you describe how you design the interactive dashboard.

(1) Select one of datasets

Select one of two datasets provided bellow (you can download the two datasets from Moodle):

a) assessment task 4-data-1-school-info.csv
b) assessment task 4-data-2-order-info.xlsx

(2) Create a plan for your visualization

After selecting dataset, you should create a design plan(proposal) in which you describe how you design an interactive dashboard based on the data. Starting by understanding and analyzing your elected data and then plan what tasks and the techniques are a good match for one another. In addition, you should also describe why the interaction techniques you will implement will be effective in the context of your tasks and carefully plan the various ideas that go into the software implementation. This is an important task that you perform first before creating an interactive dashboard. The following steps are for your reference.

(3) Data analysis and abstraction

Based on the selected data, you may further analyse and understand the data. For example, new attributes may need to be derived from existing data, or existing data may be transformed from one type into another, etc.

- Do not just draw what you are given; decide what the right thing to show is; create it with a series of transformations from the original dataset, and draw that!
- Deriving new data attributes is a critical part of the visualisation design process. The common case is that deriving new data is a choice made by vis designers.

(4) Describe scenarios based on your dataset

A scenario refers to what a user would have to do and what he or she would see step-by-step in performing a task using your visual project. The key distinction between a scenario and a task is that a scenario is design-specific, in that it shows how a task would be performed if you adopt a particular design; while the task itself is design-independent. For example, you may think about who would be interested in understanding this data better; what these people would want to know about the data, and what the problems are being addressed, etc.

(5) Identify your questions and tasks

Using the previous business scenario as a starting point in your design plan, you should make list of at least 5 questions that might be asked from your choice of dataset. In addition, you should define the tasks that you want to complete and present an appropriate design for those tasks. For example, you describe your visualizations with various charts to be used, present intended tasks, and provide the rationale for why the designs address the tasks, etc. You should also avoid asking questions that could be answered with a single statistic, such as only asking min, max, etc. Instead, your visualization should tell **an interesting story** in which your visualization adds value.

(6) Describe your proposed implementation approach.

In this project, you are required to create an interactive dashboard which consists of at least four worksheets. For each visualization, you should describe what the story you want to tell is and why the design you select such as charts is appropriate. Therefore, you need to pick up your visualization types based on the kinds of questions you are answering.

In your plan, you need to apply the visual design principles discussed in class so far to design the dashboard. For example, you should focus on what kinds of the abstract tasks you want to support and your selected encoding approaches should best support your tasks. Finally, you should sketch your solutions.

(7) Your design plan should include at least three pages (word count: 800)

2. Implement your design with analysis report (35%)

In the implementation of your design, you will put together everything you have learned in this course to create a story-based interactive dashboard to explore the dataset you selected. Based on the plan you created in the previous step, your final

capstone project should create a dashboard that includes two main components: Implement the design by using software and provide an analysis report.

(1) Implementation of your design:

Your implementation of design should include the following parts:

1. A minimum of four worksheets that are displayed on dashboard's panels
2. Creating several calculated fields to express how much, how many, ratio, comparison, etc.
3. Creating parameter fields to control selected attributes in your data
4. Adding interactive techniques into your worksheets such as filter and action filters, highlights, etc.
5. Demonstrating that you have grasped the important concepts and techniques associated with Tableau
6. Drilling through, paging or cascading techniques are also encouraged to show summary and detail views
7. A map component
8. Exploring different encodings (e.g., use colour and size - in addition to the basic chart types)

(2) Written analysis report of implementation (1200 words)

The written analysis report should include the following parts:

1. Executive summary of your visualization.
2. Your written report should list at least 5 questions with related answers, along with a screenshot showing the visualization you used to answer each question. One page per question – screenshot and narrative. Each question should be answered with a different visualization.
3. For each view, describe your visual encoding choices and include the rationale for your design choices and explain how your design and implementation answer the questions you ask.
4. Your analyzing reports must include related screenshots that are created by yourself.
5. Including a usage scenario walking through how your interactive dashboard can be used during an interactive session, illustrated with screenshots of your dashboard in action.
6. Providing a rigorous rationale for your design decisions. Document the visual encodings (e.g., calculated fields, parameters, etc.) you used to arrange the data and discuss whether those choices are appropriate for the data. How did your decision of which visualizations were suitable to use in your story facilitate effective communication?
7. In the part of reflection in the analysis report, discuss how you have created your visualization from your initial plan. What lessons do you learn from

designing and implementing this assessment task? Give an honest assessment of your project and discuss strengths, weaknesses, and limitations of your visualization. What are challenges when developing this project?

8. Implementation of your design may be different than what you originally planned in your proposal. Explain why it may be and describe changes between the initial plan and the final implementation. How have your visualization goals changed?
9. Brief introduction to the background research carried out and the course materials applied in the assessment task
10. Conclusions/findings, together with any recommendations.
11. Any reference you may use.

3. Software

You must use software Tableau or D3.js to implement the interactive dashboard. If you use D3 to implement your visualization, you must submit your codes.

4. Marking criteria

(1) Marking guide for design plan

Marking Criteria	Marks
Analyse Data • data transformation using calculated field or formula • explanation on data types • data analysis in terms of data grouping, filter, relationship with targeted audience	**/2**
Create Creative Questions • multi-variable, derived field, at least five questions • interestingness of the questions and answer planning • explanation of the questions with related data	**/3**
Describe scenario • planning and building scenario for story • unfolding the story in association with data • relating the questions and answers within story	**/2**
Design of visuals • connecting the visualisation plan with question and story of portrayed scenario • explanation of visualisation methods to communicate the answers of the questions or knowledge of the data	**/3**
Overall proposed implementation approach • visual story telling planning (exploratory analysis) • the explanation of implementation plan (effectiveness) • evidence of using/considering design principles	**/3**
Design plan structure and the plan readability • plan writing - grammar, spelling, reference • plan writing layout and organisation • providing answer, conclusion on the questions • ability to adhere to constraint of word count • plan includes a cover letter including your name and ID number	**/2**
Total Marks	**/15**

(2) Marking guide for implementation of design plan with analysis report

Marking Criteria	Marks
Create minimum 4 worksheets • Apply design principles on each worksheet • Application of designing dashboard principles in orientating worksheets • Effective use of space in worksheet and panels • Use of filter, parameter, calculated field in the worksheets	**4**
Visualise effective representation of the data • Complexity of the data (e.g., number of variables, derived data, trend, comparative, divergence etc.) used (in at least two worksheets) • Effective use of channel controls to convey complicated data • Practicability of visualisation (charts) choices supporting complex data (alignment to design principles).	**5**
Create interactive dashboard • Effective use of interaction to focus attention (main story/question) • Effective use of dashboard space used for interactivity (aesthetics) • The use of interactivity provides enhanced understanding (does the interactivity answer to the main question and subsequent questions?) • Multiple uses of interactivity (filter, annotations, drilldown, highlights, and Map navigation)	**7**
Implement Visuals (Visual story telling) • Documentation of visual encodings (e.g., calculated fields, choice of filters, parameters, etc.) used to arrange, slice the data in relation to questions asked • Discussion of encoding choices regarding appropriateness for the data • Evidence of using different kind of visualisation methods (4 more) including a geo-spatial interaction	**7**
Overall display, the visualization supports different analytical questions about the data • Level of ease in comprehending the interaction and answer to the questions • Effectiveness of interactivity with/between panels to assist communicating the scenario answer(s) • Combination of vis appropriate – i.e., choosing appropriate media, appropriate context for the data • Evidence of using/considering design principles (channels, markers etc.)	**6**
Analysis report readability • Effective use of question summary, answer, and conclusion • Standard of academic writing (structure, reference, grammar etc.) • Contains appropriate screenshot(s) showing the visualisation used from Dashboard • Concisely describes the scenario and any applicable background research • Critical reflection of analysis process, limitations, and Tableau exploration and analysis • Report includes a cover letter including your name and ID number • Ability to adhere to constraint of word count	**6**
Total Marks	**/35**

5. Late penalty:

Late assignments are highly discouraged. A penalty of 5% per day (including weekends and public holidays) will be assessed on late assignments. Under NO circumstances will assignments more than 7 days late be accepted(the assessment will be scored at zero).

Feedback

Feedback will be given via Moodle. Marks are in fdlGrades and Moodle, and individual feedback considering each assessment criterion is provided via Moodle.

Academic misconduct

To submit your assessment task, you must indicate that you have read and understood, and comply with, the Federation University Australia Academic Integrity and Student Plagiarism policies and procedures (http://policy.federation.edu.au/learning_and_teaching/compliance/academic_integrity/ch02.php).

You must also agree that your work has not been outsourced and is entirely your own except where work quoted is duly acknowledged. Additionally, you must agree that your work has not been submitted for assessment in any other course or program.